

99

1) The individual layers in a TRISO particle and their purposes are listed below:

10/10

- Fuel Kernel: There is flexibility in the type of fuel used (e.g., UO_2 , UCO , ThO_2 , etc.). The purpose is to allow for fission to occur.
- Inner Pyrolytic Carbon (IPyC): The purpose of the IPyC layer is to put the ^{internal} SiC into compression and protect the SiC from fission product attack.
- Silicon Carbide (SiC): The purpose of the SiC is to be the pressure vessel and primary fission product (FP) barrier.
- Outer Pyrolytic Carbon (OPyC): The purpose of the OPyC is to protect the SiC ^{layer} during ~~manufacturing~~ fabrication, and put the SiC under compression.
- Buffer: The buffer is a porous carbon layer that provides space for fuel kernel swelling and fission gases. It also absorbs fission recoil.

2) The trade-offs between carbide and oxide fuel kernels include carbide fuel kernels ~~being~~ performing better at high burnups and high power densities compared to oxide fuel. Fission of oxide fuel leads to excess oxygen being generated, which can react with the buffer and Pyc to form CO and CO₂. ~~Primarily~~ CO is a key concern as it increases pressure in the TRISO particle and adds stress to the structural layers; this could eventually result in the TRISO particle failing if the SiC cracks. Oxide fuel (UO₂) retains fission products better than carbide fuel. Carbide fuel has the benefit of not generating excess oxygen during fission. Therefore, CO generation and overpressurization is not a major concern for carbide fuels ~~unlike~~ unlike UO₂. The U.S. has pursued UCO (i.e., a combination of UC₂ and UO₂) so the UC₂ can get the excess oxygen resulting from fission and prevent a lot of CO from being produced. UO₂ kernels can also have concerns of the amoeba effect.

3)

Crystal Bower

8/8

Regarding fission product chemistry in the Fuel Kernel, in UO_2 Kernels, oxygen (O) oxidizes rare earth fission products as they're soluble and form rare earth oxides. In UO_2 Kernels, Zr reacts with O to form ZrO_2 and Intermetallic Precipitates (e.g., Pd) form, which can be released from the Kernel and attack the SiC layer. In Carbide fuels like UCO , O will react with the UC_2 phase first, then rare earth fission products (FPs). Carbide compounds form. As burnup increases, the amount of the UC_2 phase decreases. Eventually, the Oxygen potential ~~decreases~~ will increase and be able to form oxides and react with other fission products once the UC_2 phase is depleted. ~~decreases~~ In nuclear fuel, there are five main FP types: Soluble Oxides (e.g., Y), Insoluble Oxides, metals, Volatiles, and Noble gases. The Oxygen potential is the partial pressure of O in the gas phase in the fuel; ~~from~~ the oxygen potential enables you to know which elements in their competition for O , will form oxides. A higher O potential means more CO in UO_2 Kernels.

4)

Crystal Boundaries

18/
18

At low temperatures ($< 400\text{K}$), black spot defects (likely self-interstitial atom clusters) characterize the microstructure of SiC. Strain accumulation occurs due to irradiation-induced defects until it reaches a critical amorphization limit at which the crystal becomes amorphous. SiC swells logarithmically with dose until the ^{critical} amorphization limit at 423K .

At low to medium temperatures ($423\text{K} - 1273\text{K}$), black spot defects dominate and this is called the point defect swelling regime. SiC swells ~~logarithmically~~ with dose until it reaches a saturation level. The saturation level decreases with an increase in temperature due to defect recombination.

At high temperatures (1000°C), interstitial, sessile Frank loops and voids form ^{leading to swelling}. Vacancies are mobile and vacancy clusters form. This is the void swelling regime. There's no known saturation level for void swelling.

5) Thermal conductivity (K_{th}) changes in SiC
 with irradiation by at low temperatures,
~~simple defects form like~~
 self-interstitial atom clusters / black dot defects
 which act as the primary phonon scatterers.
 K_{th} degrades until it reaches saturation
 saturation within this point defect
 swelling regime (usually a few dpa).
 The thermal defect resistance is $\frac{1}{K_{TDR}} = \frac{1}{K_{TDC}} - \frac{1}{K_{unirr}}$
 and is proportional to swelling, so you
 can measure the density change of the
 SiC layer as an indirect way to determine
 the thermal conductivity (K_{th}). In ^{in the point defect swelling regime only.}
 the void swelling regime, K_{th} degrades
 as voids grow; ~~however~~
~~however~~
 K_{th} degrades less in the void
 swelling regime than the
 point defect swelling
 regime.

18/18
6) Graphite dimensionally changes under irradiation by interstitials forming a bridge-like structure between basal planes and repelling the planes leading to an extension in the c direction and contraction in the a direction. Vacancies on the basal plane also lead to a contraction in a. Graphite initially shrinks under irradiation, then reaches a turnaround point and swells. This is because porosity and Mrozowski cracks will close and accommodate the ^{expansion} ~~extension~~ in the c direction so the overall, net change is shrinking. Once porosity is closed, the expansion in c dominates leading to swelling. Mrozowski cracks are thin, micro cracks parallel to the basal planes.

7)

9/9

Three types of failure mechanisms for the TRISO particles are listed and described below:

- Over pressurization occurs as fission gas is generated in the fuel kernel and applies force to the structural layers (especially if the buffer layer ~~was~~ was damaged or deformed during fabrication). CO and CO₂ can form from excess oxygen being generated in UO₂ kernels ^{during fission} and interacting with the buffer and PyC. CO and CO₂ add to the pressure in the TRISO particle. Overpressurization can lead to TRISO particle failure as the SiC fails.
- Fission Product Attack: Fission product attack occurs as fission products are generated during fission. ^{as} Palladium is a key fission product concern as it can attack the inner SiC layer and Pd-SiC compounds can consume the SiC layer leading to SiC, and ultimately TRISO ~~when~~ particle failure.

7 (cont.)

Fission products can chemically react with SiC and lead to its failure. Fission products form in the fuel kernel ~~in the fuel kernel~~ and can travel to the other TRISO coatings.

• Inner Pyrolytic Carbon (IPYC) - SiC

Debonding: A weak bond between the IPYC and SiC layer during fabrication can lead to excess tensile stress being applied to SiC as the IPYC shrinks. This leads to the SiC, ~~and TRISO~~ and therefore, the TRISO particle failing.

• Kernel Migration / Amoeba Effect:

The fuel kernel migrates toward the TRISO coatings resulting in TRISO particle failure because carbon migrates up the thermal gradient, while the UO_2 kernel migrates down the thermal gradient.

8) One example of an advanced TRISO concept is replacing the SiC layer with zirconium carbide (ZrC).

3/3

This is of interest because ZrC has improved fission product retention, fission product attack mitigation, and thermal stability capabilities. SiC undergoes thermal decomposition when ~~temperatures~~ temperatures exceed 2000°C but ZrC has been tested and did not fail up to 2400°C .

~~However, ZrC can~~ ZrC can help address Pd (Palladium) attack concerns that SiC faces; however, the silver (Ag) problem still continues with it passing through intact layers likely by grain boundary diffusion.

9) Concerns for intermediate heat exchanger materials in a He environment includes corrosion and environmental degradation. Intermediate heat exchangers (IHXs) have to withstand high operating temperatures and long service lives.

The He environment where He ~~coolant~~ is the coolant includes impurities (e.g., H_2O , N_2 , etc.) that can lead

Carburization
+
He carburization?

to corrosion of the IHX. Creep can also be a concern at these high temperatures. Therefore, candidate materials include nickel-based alloys and Alloy 617. A chromia layer / oxide layer is preferred for the material to combat environmental corrosion. Solid-solutioning carbides are added internally to the IHX material to improve strength and creep resistance. Systems to clean the gas (He) are added to the plant to remove impurities that present corrosion concerns to the IHX and other plant components. Alloy 617 is only qualified to 750°C currently but higher temperatures are expected.